



Brief paper

Adaptive distributed observer for general linear leader systems over periodic switching digraphs[☆]Changran He^a, Jie Huang^{a,b,*}^a Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Shatin, N.T., Hong Kong^b Shenzhen Research Institute, The Chinese University of Hong Kong, Hong Kong

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ABSTRACT

A standard assumption on the existence of the distributed observer or adaptive distributed observer for a linear leader system over jointly connected switching networks is that none of the eigenvalues of the system matrix of the leader system have positive real parts. In this paper, after establishing a stability result for a class of linear periodic switched systems, we present sufficient conditions for the existence of both the distributed observer and the adaptive distributed observer for general linear leader systems over periodic jointly connected switching networks. As an application of the distributed observer, we solve the cooperative output regulation problem for linear multi-agent systems with general linear leader systems over periodic jointly connected switching networks. A numerical example is provided to illustrate our design.

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1. Introduction

The distributed observer is an effective tool to handle various leader-following control problems of multi-agent systems, such as the consensus problem (Hong & Wang, 2009; Jadbabaie, Lin, & Morse, 2003; Olfati-Saber & Murray, 2004; Zhang, Xu, Wang, & Zhao, 2017), the cooperative output regulation problem (Li, Feng, Wang, Luo, & Guan, 2015; Su & Huang, 2012a, 2012b), the leader-following output synchronization problem (Yang, Modares, Wunsch, & Yin, 2018; Zhu, Chen, & Middleton, 2016), and the rendezvous problem (Dong & Chen, 2019). A distributed observer for a leader system is a distributed dynamic compensator that can provide for each follower the estimated state of the leader system. The distributed observer was first developed for a general linear leader system over static communication networks in Su and Huang (2012a), and then it was extended to handle a linear

leader system over jointly connected switching communication networks in Su and Huang (2012b). However, the distributed observers in Su and Huang (2012a, 2012b) have a drawback that every follower must know the system matrix of the leader system. To overcome this drawback, the so-called adaptive distributed observer was proposed in Cai and Huang (2016), which can estimate both the state and the system matrix of a neutrally stable linear leader system over jointly connected switching networks, and thus it only requires the children of the leader to know the system matrix of the leader system.

Nevertheless, the distributed observer over jointly connected switching networks studied in Su and Huang (2012b) or the adaptive distributed observer over jointly connected switching networks studied in Cai and Huang (2016) only applies to linear leader systems whose system matrices do not have eigenvalues with positive real parts. Thus, there is a big gap between the result of the distributed observer over static communication networks and the result of the distributed observer over jointly connected switching communication networks. It is interesting to further consider general linear leaders over jointly connected switching networks.

In this paper, we will explore the possibility of establishing the adaptive distributed observer, which includes the distributed observer as a special case, for a general linear leader system over jointly connected switching communication networks. By first establishing a stability result for a class of linear periodic switched systems, we manage to show that an adaptive distributed observer for a general linear leader system over jointly connected switching communication networks indeed exists

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provided that the switching signal is periodic with its period sufficiently small. Our main result is also applied to the cooperative output regulation of linear multi-agent systems with general leader systems over periodic jointly connected switching networks. It is noted that the cooperative output regulation of linear multi-agent systems with general leader systems over periodic switching networks was also studied in Li et al. (2015). However, in Li et al. (2015), the agents are divided into two groups: exosystem-accessible agents and other agents. Two more restrictions are imposed on the second group of agents in Li et al. (2015). First, the communications among the second group of agents are undirected. Second, the second group of agents is every-time connected to some agent in the first group. In contrast, our jointly connected switching networks can be directed and disconnected at every time instant, which makes our problem much more challenging.

It is worth noting that the existence of the distributed observer for a linear leader system is closely related to the solvability of the leader-following consensus problem for linear multi-agent systems. So far, the consensus problem for linear multi-agent systems over static networks has been thoroughly studied in Hu and Hong (2007), Tuna (2008), Zhang, Lewis, and Das (2011), Zhu and Jiang (2015) and Zhao and Lin (2016), to just name a few. However, the solvability of the leader-following consensus problem for linear multi-agent systems over jointly connected switching networks also relies on the restrictive condition that none of the eigenvalues of the system matrix of each agent have positive real parts (Jadbabaie et al., 2003; Ni & Cheng, 2010; Su & Huang, 2012c). Thus, there is also a big gap between the result of the linear leader-following consensus problem over static communication networks and the result of the linear leader-following consensus problem over jointly connected switching communication networks. By narrowing this gap, the result of this paper has also significantly advanced the status of the linear leader-following consensus problem over jointly connected switching networks.

The rest of this paper is organized as follows. In Section 2, we establish a stability result for a class of linear periodic switched systems and propose an adaptive distributed observer for general linear leader systems over periodic switching digraphs satisfying the jointly connected condition. In Section 3, we apply the above result to solve the cooperative output regulation problem of linear multi-agent systems. A numerical example is provided to illustrate our design in Section 4. The paper is closed in Section 5 with some concluding remarks. A preliminary version of this paper appeared in the conference paper (He & Huang, 2020).

Throughout this paper, we adopt the following notation: $\mathbb{R}$ denotes the set of real numbers. $\mathbf{1}_N$ denotes the N -dimensional column vector with all elements 1. For $A_i \in \mathbb{R}^{p_i \times m}$, $i = 1, \dots, n$, $\text{col}(A_1, \dots, A_n) = [A_1^T \ \dots \ A_n^T]^T$.

2. Main results

2.1. A stability result

Consider a switching digraph¹ $\mathcal{G}_{\sigma(t)} = (\mathcal{V}, \mathcal{E}_{\sigma(t)})$ with $\mathcal{V} = \{0, 1, \dots, N\}$ and $\mathcal{E}_{\sigma(t)} \subseteq \mathcal{V} \times \mathcal{V}$ for all $t \geq 0$. Assume that the switching digraph $\mathcal{G}_{\sigma(t)}$ satisfies the following assumptions.

Assumption 2.1. There exists a subsequence $\{i_k \mid k = 0, 1, 2, \dots\}$ of $\{i \mid i = 0, 1, 2, \dots\}$ with $t_{i_{k+1}} - t_{i_k} < \nu$ for some $\nu > 0$, such that the union graph $\bigcup_{j=i_k}^{i_{k+1}-1} \mathcal{G}_{\sigma(t_j)}$ has the property that there is a path from node 0 to every other node.

Assumption 2.2. The switching signal $\sigma(t)$ is periodic with period T .

Remark 2.1. Assumption 2.1 is called the jointly connected condition, and it is the mildest assumption on switching networks since it allows the graph to be disconnected at any time instant. Under Assumptions 2.1 and 2.2, without loss of generality, we can assume that the switching signal $\sigma(t)$ is of the following form:

$$\sigma(t) = \begin{cases} 1, & \text{if } sT \leq t < (s + \omega_1)T \\ 2, & \text{if } (s + \omega_1)T \leq t < (s + \sum_{p=1}^2 \omega_p)T \\ \vdots & \\ \rho, & \text{if } (s + \sum_{p=1}^{\rho-1} \omega_p)T \leq t < (s + 1)T \end{cases} \quad (1)$$

where $s = 0, 1, 2, \dots$, and ω_p , $p = 1, \dots, \rho$, are positive constants satisfying $\sum_{p=1}^{\rho} \omega_p = 1$. Then, for any $k = 0, 1, 2, \dots$, the union graph $\bigcup_{j=i_k}^{i_{k+1}-1} \mathcal{G}_{\sigma(t_j)} = \bigcup_{p=1}^{\rho} \mathcal{G}_p$. A switching digraph $\mathcal{G}_{\sigma(t)}$ satisfying Assumptions 2.1 and 2.2 is said to be periodically jointly connected.

Let us first study the stability of the following class of linear periodic switched systems:

$$\dot{\vartheta} = (I_N \otimes G - \mu(H_{\sigma(t)} \otimes I_n)) \vartheta \quad (2)$$

where $\vartheta \in \mathbb{R}^{Nn}$ is the state, $G \in \mathbb{R}^{n \times n}$ is a constant matrix, $\mu \in \mathbb{R}$ is a positive constant, and $H_{\sigma(t)} = [h_{ij}(t)]_{i,j=1}^N \in \mathbb{R}^{N \times N}$ is obtained from the Laplacian matrix $\mathcal{L}_{\sigma(t)}$ of $\mathcal{G}_{\sigma(t)}$ by removing the first row and the first column of $\mathcal{L}_{\sigma(t)}$. We start with rephrasing Lemma 3.22 of Sun and Ge (2005) as follows.

Lemma 2.1. Consider the linear switched system as follows:

$$\dot{x} = A_{\sigma(t)} x \quad (3)$$

where $x \in \mathbb{R}^n$ is the state, $\sigma : [0, \infty) \rightarrow \mathcal{P} = \{1, 2, \dots, \rho\}$ is the switching signal satisfying Assumption 2.2, and $A_p \in \mathbb{R}^{n \times n}$, $p = 1, \dots, \rho$. If the matrix $\sum_{p=1}^{\rho} \omega_p A_p$ is Hurwitz with ω_p , $p = 1, \dots, \rho$, obtained from (1), then there exists a positive constant $\bar{T}$ such that, for $0 < T < \bar{T}$, the origin of system (3) is exponentially stable.

We are now ready to obtain the stability result of system (2) as follows.

Lemma 2.2. Consider the linear switched system (2). Under Assumptions 2.1 and 2.2, there exist some constants $\bar{\mu}_0$ and $\bar{T}_0$, such that, if $\mu > \bar{\mu}_0$ and $0 < T < \bar{T}_0$, then the origin of system (2) is exponentially stable.

Proof. Let us first show that the matrix $-\sum_{p=1}^{\rho} \omega_p H_p$ is Hurwitz. For this purpose, denote the Laplacian matrix of the digraph $\mathcal{G}_p$ by $\mathcal{L}_p$. Then,

$$\mathcal{L}_p = \left[\begin{array}{c|c} l_{00}^p & l_0^p \\ \hline \theta_p & H_p \end{array} \right], \quad p = 1, \dots, \rho \quad (4)$$

where $H_p \in \mathbb{R}^{N \times N}$, $\theta_p = (-H_p) \mathbf{1}_N \in \mathbb{R}^N$, $l_{00}^p \geq 0$, and $l_0^p \in \mathbb{R}^{1 \times N}$ whose elements are nonpositive such that $l_{00}^p + l_0^p \mathbf{1}_N = 0$. It is noted that the matrices H_p , $p = 1, \dots, \rho$, are diagonally dominant with nonnegative diagonal elements.

Next, let

$$\hat{\mathcal{L}} = \left[\begin{array}{c|c} 0_{1 \times 1} & 0_{1 \times N} \\ \hline \hat{\theta} & \sum_{p=1}^{\rho} \omega_p H_p \end{array} \right] \quad (5)$$

where $\hat{\theta} = (-\sum_{p=1}^{\rho} \omega_p H_p) \mathbf{1}_N \in \mathbb{R}^N$. Since the matrices H_p , $p = 1, \dots, \rho$, are diagonally dominant with nonnegative diagonal elements, so is $\sum_{p=1}^{\rho} \omega_p H_p$. Thus, all components of $\hat{\theta}$ are

¹ See the appendix of He and Huang (2021) for a summary of digraph.

nonpositive. Since all the off-diagonal elements of the matrix $-\hat{\mathcal{L}}$ are nonnegative, $-\hat{\mathcal{L}}$ is a Metzler matrix with zero row sum.

Let $\mathcal{G} = \bigcup_{p=1}^{\rho} \mathcal{G}_p$. Then, by Remark 2.1, under Assumptions 2.1 and 2.2, for any $k = 0, 1, 2, \dots$, the union graph $\bigcup_{j=l_k}^{i_{k+1}-1} \mathcal{G}_{\sigma(i_j)} = \bigcup_{p=1}^{\rho} \mathcal{G}_p = \mathcal{G}$. Let $\mathcal{L} = [l_{ij}]_{i,j=0}^N \in \mathbb{R}^{(N+1) \times (N+1)}$ be the Laplacian matrix of the digraph $\mathcal{G}$ and $\hat{\mathcal{G}}$ be the digraph associated with the Metzler matrix $-\hat{\mathcal{L}}$, where $\hat{\mathcal{L}} = [\hat{l}_{ij}]_{i,j=0}^N \in \mathbb{R}^{(N+1) \times (N+1)}$. Then, it can be verified that, for $i = 1, \dots, N$ and $j = 0, 1, \dots, N$, $\hat{l}_{ij} > 0$ if and only if $l_{ij} > 0$, $\hat{l}_{ij} < 0$ if and only if $l_{ij} < 0$, and $\hat{l}_{ij} = 0$ if and only if $l_{ij} = 0$. Thus, $\hat{\mathcal{G}}$ is obtained from $\mathcal{G}$ by removing all such edges as $(i, 0)$, $i = 1, \dots, N$. Under Assumptions 2.1 and 2.2, node 0 can reach all other N nodes in the digraph $\mathcal{G}$. Thus, node 0 can also reach all other N nodes in the digraph $\hat{\mathcal{G}}$. By Lemma 3.3 of Ren and Beard (2005), the matrix $\hat{\mathcal{L}}$ has one simple eigenvalue at the origin, and all other eigenvalues of $\hat{\mathcal{L}}$ have positive real parts. Thus, from the block lower triangular structure of $\hat{\mathcal{L}}$, the matrix $-\sum_{p=1}^{\rho} \omega_p H_p$ is Hurwitz.

Given any constant matrix G , the eigenvalues of the matrix $I_N \otimes G - \mu (\sum_{p=1}^{\rho} \omega_p H_p) \otimes I_n$ are given by

$$\left\{ \lambda_i(G) - \mu \lambda_j \left(\sum_{p=1}^{\rho} \omega_p H_p \right) \right\} \quad (6)$$

where $\lambda_i(G)$ and $\lambda_j(\sum_{p=1}^{\rho} \omega_p H_p)$ are the eigenvalues of G and $\sum_{p=1}^{\rho} \omega_p H_p$, respectively. Let $\bar{\lambda}_{\max}(G)$ be an eigenvalue of G with the largest real part, let $\bar{\lambda}_{\min}(\sum_{p=1}^{\rho} \omega_p H_p)$ be an eigenvalue of $\sum_{p=1}^{\rho} \omega_p H_p$ with the smallest (positive) real part, and let $\text{Re}(\cdot)$ be the real part of a complex number. Let

$$\bar{\mu}_0 = \frac{\text{Re}(\bar{\lambda}_{\max}(G))}{\text{Re}(\bar{\lambda}_{\min}(\sum_{p=1}^{\rho} \omega_p H_p))}. \quad (7)$$

Then, for any positive constant $\mu > \bar{\mu}_0$, the matrix $I_N \otimes G - \mu (\sum_{p=1}^{\rho} \omega_p H_p) \otimes I_n$ is Hurwitz. Since $\sum_{p=1}^{\rho} \omega_p = 1$, $I_N \otimes G - \mu (\sum_{p=1}^{\rho} \omega_p H_p) \otimes I_n = \sum_{p=1}^{\rho} \omega_p (I_N \otimes G - \mu (H_p \otimes I_n))$. Invoking Lemma 2.1 with $A_p = I_N \otimes G - \mu (H_p \otimes I_n)$ and $\mu > \bar{\mu}_0$ completes the proof. $\square$

2.2. An estimate of $\bar{T}$

Since Lemma 2.1 did not provide an estimate of the positive constant $\bar{T}$, it is desirable to give an estimate of $\bar{T}$. For this purpose, let us first rephrase Lemma 1 of Tokarzowski (1987) as follows.

Lemma 2.3. Let $A_p \in \mathbb{R}^{n \times n}$, $p = 1, \dots, \rho$, $\alpha = \max\{\|A_p\| \mid p = 1, \dots, \rho\}$, and ω_p , $p = 1, \dots, \rho$, be positive constants satisfying $\sum_{p=1}^{\rho} \omega_p = 1$. If $\bar{T} > 0$ is such that

$$\sum_{p=1}^{\rho} \left(\alpha \bar{T} + \frac{1}{2} \alpha^2 \bar{T}^2 \exp(\alpha \bar{T}) \right)^p < 1 \quad (8)$$

then, for all $0 < T \leq \bar{T}$, we have

$$\prod_{p=1}^{\rho} \exp(\omega_p A_p T) = \exp \left(\left(\sum_{p=1}^{\rho} \omega_p A_p + O(T) \right) T \right) \quad (9)$$

where $\prod_{p=1}^{\rho} \exp(\omega_p A_p T) = \exp(\omega_{\rho} A_{\rho} T) \dots \exp(\omega_1 A_1 T)$ and $O(T) \in \mathbb{R}^{n \times n}$ is some matrix function of T such that

$$\|O(T)\| \leq R_{\rho}(T) \quad (10)$$

with

$$R_{\rho}(T) = \frac{1}{2} \alpha^2 T \exp(\alpha T)$$

$$+ \frac{1}{T} \sum_{p=2}^{\rho} \left(\alpha T + \frac{1}{2} \alpha^2 T^2 \exp(\alpha T) \right)^p + \frac{1}{2T} \frac{\left(\sum_{p=1}^{\rho} \left(\alpha T + \frac{1}{2} \alpha^2 T^2 \exp(\alpha T) \right)^p \right)^2}{1 - \sum_{p=1}^{\rho} \left(\alpha T + \frac{1}{2} \alpha^2 T^2 \exp(\alpha T) \right)^p} \quad (11)$$

which is strictly increasing for all $0 < T \leq \bar{T}$.

Let us also rephrase Theorem 7.2.66 of Callier and Desoer (1991) as follows.

Lemma 2.4. Consider the following linear periodic system:

$$\dot{x} = A(t)x \quad (12)$$

where $x \in \mathbb{R}^n$ and $A : [0, \infty) \rightarrow \mathbb{R}^{n \times n}$ is piecewise continuous and T -periodic, i.e.,

$$A(t + T) = A(t), \quad \forall t \geq 0. \quad (13)$$

Then, the origin of system (12) is exponentially stable if and only if its state transition matrix $\Phi(T, 0)$ is Schur.

Lemma 2.5. Suppose system (3) satisfies Assumption 2.2, $A = \sum_{p=1}^{\rho} \omega_p A_p$ is Hurwitz, and P is the symmetric and positive definite matrix satisfying $A^T P + PA = -I$. If $\bar{T} > 0$ satisfies

$$\sum_{p=1}^{\rho} \left(\alpha \bar{T} + \frac{1}{2} \alpha^2 \bar{T}^2 \exp(\alpha \bar{T}) \right)^p < 1 \quad (14)$$

where $\alpha = \max\{\|A_p\| \mid p = 1, \dots, \rho\}$, and

$$R_{\rho}(\bar{T}) \leq \frac{1}{2\lambda_{\max}(P)} \quad (15)$$

where $R_{\rho}(\cdot)$ is as defined in (11) and $\lambda_{\max}(P)$ is the largest eigenvalue of P , then system (3) is exponentially stable for all $0 < T < \bar{T}$.

Proof. Denote the state transition matrix of (3) by $\Phi(t, t_0)$. Then it can be verified that $\Phi(T, 0) = \prod_{p=1}^{\rho} \exp(\omega_p A_p T)$. By Lemma 2.4, system (3) is exponentially stable if and only if $\Phi(T, 0)$ is Schur. Since (14) is satisfied, by Lemma 2.3, for $0 < T \leq \bar{T}$, we have

$$\begin{aligned} \Phi(T, 0) &= \exp \left(\left(\sum_{p=1}^{\rho} \omega_p A_p + O(T) \right) T \right) \\ &= \exp((A + O(T))T) \end{aligned} \quad (16)$$

and $\|O(T)\| \leq R_{\rho}(T)$. Thus, by Lemma 2.4, system (3) is exponentially stable for all $0 < T < \bar{T}$ if and only if $A + O(T)$ is Hurwitz for all $0 < T < \bar{T}$.

Consider the linear system

$$\dot{x} = (A + O(T))x. \quad (17)$$

Let $V(x) = x^T P x$ be a Lyapunov function candidate for (17). Since $R_{\rho}(T)$ is strictly increasing for all $0 < T \leq \bar{T}$, for $0 < T < \bar{T}$, we have

$$\begin{aligned} \dot{V} &= x^T ((A + O(T))^T P + P(A + O(T)))x \\ &= -x^T x + 2x^T P O(T)x \\ &< -\|x\|^2 + 2\lambda_{\max}(P)R_{\rho}(\bar{T})\|x\|^2, \quad \forall x \neq 0. \end{aligned} \quad (18)$$

Thus, in view of (15), $A + O(T)$ is Hurwitz for all $0 < T < \bar{T}$. $\square$

Remark 2.2. The estimate of $\bar{T}$ given by Lemma 2.5 may be conservative. If the period T of system (3) is known, then it is easier to directly check the location of the eigenvalues of the matrix $\Phi(T, 0)$ as suggested in Lemma 2.4.

2.3. An adaptive distributed observer

Consider a general linear leader system of the following form:

$$\dot{v} = Sv \quad (19)$$

where $v \in \mathbb{R}^q$ and $S \in \mathbb{R}^{q \times q}$ is a constant matrix.

Given the leader system (19) and a switching digraph $\mathcal{G}_{\sigma(t)}$, Su and Huang (2012b) proposed a distributed dynamic compensator as follows:

$$\dot{\eta}_i = S\eta_i + \mu_1 \sum_{j=0}^N a_{ij}(t)(\eta_j - \eta_i), \quad i = 1, \dots, N \quad (20)$$

where $\eta_0 = v$, for $i = 1, \dots, N$, $\eta_i \in \mathbb{R}^q$, $a_{ij}(t)$ is the element of the weighted adjacency matrix of $\mathcal{G}_{\sigma(t)}$, and μ_1 is a positive constant to be determined. It was shown in Su and Huang (2012b) that, under Assumption 2.1 and the assumption that none of the eigenvalues of S have positive real parts, for any initial conditions $v(0)$, $\eta_i(0)$, $i = 1, \dots, N$, and any $\mu_1 > 0$, the solutions of systems (19) and (20) exist for all $t \geq 0$ and satisfy

$$\lim_{t \rightarrow \infty} (\eta_i(t) - v(t)) = 0, \quad i = 1, \dots, N \quad (21)$$

exponentially. For this reason, system (20) is called a distributed observer for the leader system (19).

The assumption that none of the eigenvalues of S have positive real parts is restrictive. By Lemma 2.2, it is possible to remove this assumption for periodic jointly connected switching digraphs. In fact, let $\tilde{\eta}_i = \eta_i - v$, $i = 1, \dots, N$, $\tilde{\eta} = \text{col}(\tilde{\eta}_1, \dots, \tilde{\eta}_N)$, and $\eta = \text{col}(\eta_1, \dots, \eta_N)$. From (19) and (20), we have

$$\begin{aligned} \dot{\tilde{\eta}} &= \dot{\eta} - \mathbf{1}_N \otimes \dot{v} \\ &= (I_N \otimes S - \mu_1(H_{\sigma(t)} \otimes I_q)) \eta \\ &\quad + \mu_1(H_{\sigma(t)} \otimes I_q)(\mathbf{1}_N \otimes v) - (I_N \otimes S)(\mathbf{1}_N \otimes v) \\ &= (I_N \otimes S - \mu_1(H_{\sigma(t)} \otimes I_q))(\eta - (\mathbf{1}_N \otimes v)) \\ &= (I_N \otimes S - \mu_1(H_{\sigma(t)} \otimes I_q))\tilde{\eta}. \end{aligned} \quad (22)$$

Since system (22) is in the form of (2) with $G = S$, $\mu = \mu_1$, and $\vartheta = \tilde{\eta}$, by Lemma 2.2, we have the following result.

Lemma 2.6. Under Assumptions 2.1 and 2.2, there exist positive constants $\bar{\mu}_1$ and $\bar{T}_1$, such that, if $\mu_1 > \bar{\mu}_1$ and $0 < T < \bar{T}_1$, then the origin of system (22) is exponentially stable.

Since a distributed observer of the form (20) relies on the assumption that every follower knows the system matrix S of the leader system, it is more practical to consider the case where only the children of the leader have access to the knowledge of S . Indeed, Cai and Huang (2016) further proposed a distributed dynamic compensator as follows:

$$\dot{S}_i = \mu_1 \sum_{j=0}^N a_{ij}(t)(S_j - S_i) \quad (23a)$$

$$\dot{v}_i = S_i v_i + \mu_2 \sum_{j=0}^N a_{ij}(t)(v_j - v_i), \quad i = 1, \dots, N \quad (23b)$$

where $S_0 = S$, $v_0 = v$, μ_1 and μ_2 are some positive constants to be specified, and, for $i = 1, \dots, N$, $S_i \in \mathbb{R}^{q \times q}$, $v_i \in \mathbb{R}^q$. It was shown in Liu and Huang (2019) that, under Assumption 2.1 and the assumption that none of the eigenvalues of S have positive real parts, for any initial conditions $v(0)$, $S_i(0)$, $v_i(0)$, $i = 1, \dots, N$, and any $\mu_1, \mu_2 > 0$, the solutions of systems (19) and (23) exist for all $t \geq 0$ and satisfy

$$\lim_{t \rightarrow \infty} (S_i(t) - S) = 0, \quad i = 1, \dots, N \quad (24)$$

exponentially, and

$$\lim_{t \rightarrow \infty} (v_i(t) - v(t)) = 0, \quad i = 1, \dots, N \quad (25)$$

exponentially. For this reason, system (23) is called an adaptive distributed observer for the leader system (19).

Remark 2.3. Consider the following multi-agent system:

$$\dot{\eta}_i = S\eta_i + Bu_i, \quad i = 1, \dots, N \quad (26)$$

where $B \in \mathbb{R}^{q \times m}$ for some integer $m > 0$, and the leader system (19). The leader-following consensus problem of (19) and (26) is to design a distributed control law such that (21) is achieved. Suppose $B = I_q$. Then, under the following control law:

$$u_i = \mu_1 \sum_{j=0}^N a_{ij}(t)(\eta_j - \eta_i), \quad i = 1, \dots, N \quad (27)$$

the closed-loop system is given by (20). Recall that the solvability of the leader-following consensus problem for linear multi-agent systems over jointly connected switching networks either relies on the restrictive condition that none of the eigenvalues of the system matrix of each agent have positive real parts (Jadbabaie et al., 2003; Ni & Cheng, 2010; Su & Huang, 2012c) or on the assumption that the graph is acyclic (He & Huang, 2021). Hence, it is desirable to remove these two restrictive assumptions. For this purpose, it is noted that Lemma 2.6 does not rely on these two restrictive assumptions and implies the solvability of the leader-following consensus problem for systems (19) and (26) with $B = I_q$. Thus, for the case where $B = I_q$, Lemma 2.6 has also extended the solvability of the leader-following consensus problem for systems (19) and (26) over jointly connected switching networks provided that the switching signal is periodic with its period sufficiently small.

In what follows, we will further remove the assumption that none of the eigenvalues of S have positive real parts for periodic jointly connected switching digraphs. For this purpose, let $\tilde{S}_i = S_i - S$, $\tilde{v}_i = v_i - v$, $i = 1, \dots, N$, $\tilde{S} = \text{col}(\tilde{S}_1, \dots, \tilde{S}_N)$, $\tilde{S}_d$ = block diag($\tilde{S}_1, \dots, \tilde{S}_N$), and $\tilde{v} = \text{col}(\tilde{v}_1, \dots, \tilde{v}_N)$. From (19) and (23), we have

$$\dot{\tilde{S}} = -\mu_1(H_{\sigma(t)} \otimes I_q)\tilde{S} \quad (28a)$$

$$\dot{\tilde{v}} = (I_N \otimes S - \mu_2(H_{\sigma(t)} \otimes I_q) + \tilde{S}_d)\tilde{v} + \tilde{S}_d(\mathbf{1}_N \otimes v). \quad (28b)$$

Then we obtain the following result which ascertains the existence of an adaptive distributed observer for general linear leader systems over periodic jointly connected switching digraphs.

Theorem 2.1. Under Assumptions 2.1 and 2.2, there exist positive constants $\bar{\mu}_2$, $\bar{\mu}_3$, and $\bar{T}_2$, such that, if $\mu_1 > \bar{\mu}_2$, $\mu_2 > \bar{\mu}_3$, and $0 < T < \bar{T}_2$, then, for any initial conditions $v(0)$, $S_i(0)$, and $v_i(0)$, $i = 1, \dots, N$, the solutions of systems (19) and (23) exist for all $t \geq 0$ and satisfy

$$\lim_{t \rightarrow \infty} (S_i(t) - S) = 0, \quad i = 1, \dots, N \quad (29)$$

exponentially, and

$$\lim_{t \rightarrow \infty} (v_i(t) - v(t)) = 0, \quad i = 1, \dots, N \quad (30)$$

exponentially.

Proof. For $i = 1, \dots, q$, let $\tilde{s}_i \in \mathbb{R}^{Nq}$ be the i th column of $\tilde{S}$. Then, from (28a), we have

$$\dot{\tilde{s}}_i = -\mu_1(H_{\sigma(t)} \otimes I_q)\tilde{s}_i, \quad i = 1, \dots, q. \quad (31)$$

Under [Assumptions 2.1](#) and [2.2](#), by Corollary 4 of [Su and Huang \(2012b\)](#), the origin of system (31) is exponentially stable for any $\mu_1 > 0$ and $T > 0$. Thus, $\lim_{t \rightarrow \infty} \tilde{S}_d(t) = 0$ exponentially.

In what follows, we further show that there exist positive constants $\bar{\mu}_2$ and $\bar{T}_3$, such that, for any $\mu_1 > \bar{\mu}_2$ and $0 < T < \bar{T}_3$, $\lim_{t \rightarrow \infty} \tilde{S}_d(t)(\mathbf{1}_N \otimes v(t)) = 0$ exponentially. For this purpose, let $\bar{\mu}$ be the real part of an eigenvalue of S with the largest real part and $\lambda > \bar{\mu}$ be some positive real number. Let $\bar{S} = I_q \otimes \lambda$ and $\zeta_i = (I_N \otimes e^{\bar{S}t})\tilde{s}_i$, $i = 1, \dots, q$. Then, ζ_i is governed by the following equation:

$$\begin{aligned} \dot{\zeta}_i &= (I_N \otimes (\bar{S}e^{\bar{S}t}))\tilde{s}_i + (I_N \otimes e^{\bar{S}t})\dot{\tilde{s}}_i \\ &= (I_N \otimes \bar{S})(I_N \otimes e^{\bar{S}t})\tilde{s}_i - \mu_1(I_N \otimes e^{\bar{S}t})(H_{\sigma(t)} \otimes I_q)\tilde{s}_i \\ &= (I_N \otimes \bar{S})(I_N \otimes e^{\bar{S}t})\tilde{s}_i - \mu_1(H_{\sigma(t)} \otimes I_q)(I_N \otimes e^{\bar{S}t})\tilde{s}_i \\ &= (I_N \otimes \bar{S} - \mu_1(H_{\sigma(t)} \otimes I_q))\zeta_i, \quad i = 1, \dots, q. \end{aligned} \quad (32)$$

For all $i = 1, \dots, q$, since system (32) is in the form of (22) with $S = \bar{S}$ and $\tilde{\eta} = \zeta_i$, under [Assumptions 2.1](#) and [2.2](#), by [Lemma 2.6](#), there exist positive constants $\bar{\mu}_2$ and $\bar{T}_3$, such that, for any $\mu_1 > \bar{\mu}_2$ and $0 < T < \bar{T}_3$, the origin of system (32) is exponentially stable. Thus, for $i = 1, \dots, N$,

$$\lim_{t \rightarrow \infty} \|e^{\bar{S}t}\tilde{s}_i(t)\| = \lim_{t \rightarrow \infty} e^{\lambda t}\|\tilde{s}_i(t)\| = 0 \quad (33)$$

exponentially. Since $\|e^{\bar{S}t}\| \leq \beta e^{\lambda t}$ for some positive real number β , from (33), for $i = 1, \dots, N$, we have

$$\begin{aligned} \lim_{t \rightarrow \infty} \|\tilde{S}_i(t)v(t)\| &= \lim_{t \rightarrow \infty} \|\tilde{S}_i(t)e^{\bar{S}t}v(0)\| \\ &\leq \lim_{t \rightarrow \infty} \beta e^{\lambda t}\|\tilde{S}_i(t)\|v(0) = 0 \end{aligned} \quad (34)$$

exponentially. Thus,

$$\lim_{t \rightarrow \infty} \tilde{S}_d(t)(\mathbf{1}_N \otimes v(t)) = 0 \quad (35)$$

exponentially.

Under [Assumptions 2.1](#) and [2.2](#), by [Lemma 2.6](#), there exist positive constants $\bar{\mu}_3$ and $\bar{T}_4$, such that, if $\mu_2 > \bar{\mu}_3$ and $0 < T < \bar{T}_4$, then the origin of system (28b) with $\tilde{S}_d$ and $\tilde{S}_d(\mathbf{1}_N \otimes v)$ set to zero is exponentially stable. Let $\bar{T}_2 = \min\{\bar{T}_3, \bar{T}_4\}$. The proof is completed by applying Lemma 1 of [Liu and Huang \(2018\)](#) to system (28b). $\square$

Remark 2.4. [Theorem 2.1](#) (respectively, [Lemma 2.6](#)) has extended the result in [Liu and Huang \(2019\)](#) (respectively, the result in [Su and Huang \(2012b\)](#)) from a matrix S whose eigenvalues have non-positive real parts to any matrix S provided that the switching digraph $\mathcal{G}_{\sigma(t)}$ satisfies [Assumption 2.2](#) with its period sufficiently small.

3. An application

The output regulation problem aims to design a feedback control law to achieve asymptotic tracking and disturbance rejection in the closed-loop system and the output regulation problem for linear systems has been extensively studied since the 1970s ([Davison, 1976](#); [Francis, 1977](#); [Huang, 2004](#)). The linear cooperative output regulation problem is an extension of the output regulation problem of a single linear plant to a linear multi-agent system and was studied in [Su and Huang \(2012a\)](#) and [Su and Huang \(2012b\)](#). However, a restriction in [Su and Huang \(2012b\)](#) is that the system matrix of the exosystem cannot have eigenvalues with positive real parts. In this section, by applying [Lemma 2.6](#), we remove this assumption provided that the switching signal is periodic with its period sufficiently small.

Consider the following linear multi-agent system:

$$\begin{aligned} \dot{x}_i &= A_i x_i + B_i u_i + E_i v \\ y_{mi} &= C_{mi} x_i + D_{mi} u_i + F_{mi} v \\ e_i &= C_i x_i + D_i u_i + F_i v, \quad i = 1, \dots, N \end{aligned} \quad (36)$$

where, for $i = 1, \dots, N$, $x_i \in \mathbb{R}^{n_i}$, $y_{mi} \in \mathbb{R}^{p_{mi}}$, $e_i \in \mathbb{R}^{p_i}$, and $u_i \in \mathbb{R}^{m_i}$ are the state, measurement output, error output, and input of the i th subsystem; A_i , B_i , C_i , D_i , E_i , F_i , C_{mi} , D_{mi} , and F_{mi} are constant matrices with conformable dimensions; and $v \in \mathbb{R}^q$ is the exogenous signal representing the reference signal to be tracked and the disturbance to be rejected. It is assumed that v is generated by the leader system (19).

We consider the following dynamic measurement output feedback control law from [Su and Huang \(2012b\)](#):

$$u_i = K_{1i}\xi_i + K_{2i}\eta_i \quad (37a)$$

$$\begin{aligned} \dot{\xi}_i &= A_i \xi_i + B_i u_i + E_i \eta_i \\ &\quad + L_i(C_{mi}\xi_i + D_{mi}u_i + F_{mi}\eta_i - y_{mi}) \end{aligned} \quad (37b)$$

$$\dot{\eta}_i = S\eta_i + \mu_1 \sum_{j=0}^N a_{ij}(t)(\eta_j - \eta_i), \quad i = 1, \dots, N \quad (37c)$$

where, for $i = 1, \dots, N$, $K_{1i} \in \mathbb{R}^{m_i \times n_i}$, $K_{2i} \in \mathbb{R}^{m_i \times q}$, and $L_i \in \mathbb{R}^{n_i \times p_{mi}}$ are gain matrices to be specified, $\xi_i \in \mathbb{R}^{n_i}$, $\eta_i \in \mathbb{R}^q$, $\eta_0 = v$, $a_{ij}(t)$ is the element of the weighted adjacency matrix of $\mathcal{G}_{\sigma(t)}$, and μ_1 is a positive constant to be determined.

As in [Su and Huang \(2012b\)](#), we formulate the cooperative output regulation problem as follows.

Problem 1. Given the leader system (19), the follower system (36), and a switching digraph $\mathcal{G}_{\sigma(t)}$, find a distributed control law of the form (37) such that

- (1) the origin of the closed-loop system composed of (36) and (37) with v set to zero is exponentially stable;
- (2) for any initial conditions $x_i(0)$, $\xi_i(0)$, $\eta_i(0)$, $i = 1, \dots, N$, and $v(0)$, the solution of the closed-loop system composed of (36), (37), and (19) satisfies

$$\lim_{t \rightarrow \infty} e_i(t) = 0, \quad i = 1, \dots, N. \quad (38)$$

As in [Su and Huang \(2012b\)](#), we need the following standard assumptions.

Assumption 3.1. The pair (A_i, B_i) , $i = 1, \dots, N$, is stabilizable.

Assumption 3.2. The pair (C_{mi}, A_i) , $i = 1, \dots, N$, is detectable.

Assumption 3.3. The linear matrix equations

$$\begin{aligned} X_i S &= A_i X_i + B_i U_i + E_i \\ 0 &= C_i X_i + D_i U_i + F_i \end{aligned}, \quad i = 1, \dots, N \quad (39)$$

have solution pairs (X_i, U_i) , respectively.

Remark 3.1. Under [Assumption 3.1](#), there exists a matrix $K_{1i} \in \mathbb{R}^{m_i \times n_i}$, $i = 1, \dots, N$, such that $A_i + B_i K_{1i}$ is Hurwitz. Under [Assumption 3.2](#), there exists a matrix $L_i \in \mathbb{R}^{n_i \times p_{mi}}$, $i = 1, \dots, N$, such that $A_i + L_i C_{mi}$ is Hurwitz. [Assumptions 3.1](#) to [3.3](#) are standard and necessary for cooperative output regulation, and they are also used in [Su and Huang \(2012a, 2012b\)](#). For any E_i and F_i , [Assumption 3.3](#) is satisfied if and only if, for all $\lambda \in \rho(S)$ where $\rho(S)$ is the spectrum of S , $\begin{bmatrix} A_i - \lambda I_{n_i} & B_i \\ C_i & D_i \end{bmatrix}$ has full row rank ([Huang, 2004](#)).

The solvability of [Problem 1](#) is summarized as the following theorem, whose proof is similar to that of Theorem 2 of [Su and Huang \(2012b\)](#) and is omitted due to the space limit.

Theorem 3.1. Under Assumptions 2.1 to 3.3, there exist positive real numbers $\bar{T}_1$ and $\bar{\mu}_1$ such that, for any periodic switching signal $\sigma(t)$ with its period $0 < T < \bar{T}_1$, Problem 1 is solvable by the distributed dynamic measurement output feedback control law (37) where, for all $i = 1, \dots, N$, K_{1i} is such that $A_i + B_i K_{1i}$ is Hurwitz, $K_{2i} = U_i - K_{1i} X_i$, L_i is such that $A_i + L_i C_{mi}$ is Hurwitz, and $\mu_1 > \bar{\mu}_1$.

Remark 3.2. Theorem 3.1 can be viewed as an extension of Theorem 2 of Su and Huang (2012b). In particular, the system matrix S of the leader system (19) in Theorem 3.1 can be an arbitrary square matrix, while S in Theorem 2 of Su and Huang (2012b) is assumed to have no eigenvalues with positive real parts. The price to pay for such an extension is the extra Assumption 2.2 and the constraint on the length of the period T of the switching signal $\sigma(t)$.

4. An example

As in Su and Huang (2012b), we consider the asymptotic tracking and disturbance rejection problem for the following second-order system:

$$\begin{aligned} \dot{x}_{i1} &= x_{i2} \\ \dot{x}_{i2} &= u_i + d_i \\ e_i &= \begin{bmatrix} x_{i1} \\ x_{i2} \end{bmatrix} - \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \end{aligned} \quad (40)$$

$y_{mi} = x_{i1}$, $i = 1, 2, 3, 4$.

The leader system is assumed to be

$$\dot{v} = S v = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0.5 \end{bmatrix} v \quad (41)$$

where $v = \text{col}(v_1, v_2, v_3)$. The external disturbances of the followers are assumed to be $d_i = 0.5 * i * v_3$, $i = 1, 2, 3, 4$.

The problem can be viewed as a leader-following consensus problem with an unstable leader system and four double integrators subject to exponentially growing disturbances as the followers. We note that each follower is in the form of (36) with $x_i = \text{col}(x_{i1}, x_{i2})$,

$$\begin{aligned} A_i &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B_i = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, E_i = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0.5 * i \end{bmatrix} \\ C_i &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, D_i = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, F_i = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \\ C_{mi} &= [1 \ 0], D_{mi} = 0, F_{mi} = [0 \ 0 \ 0], i = 1, 2, 3, 4. \end{aligned}$$

Then, it can be verified that Assumptions 3.1 to 3.3 are satisfied. Moreover, the solution of (39) is given by

$$X_i = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, U_i = [-1 \ 0 \ -0.5 * i], i = 1, 2, 3, 4.$$

Consider the scenario where the switching digraph $\mathcal{G}_{\sigma(t)}$ is dictated by the following periodic switching signal:

$$\sigma(t) = \begin{cases} 1, & \text{if } sT \leq t < (s+0.5)T \\ 2, & \text{if } (s+0.5)T \leq t < (s+0.8)T \\ 3, & \text{if } (s+0.8)T \leq t < (s+1)T \end{cases} \quad (42)$$

where $T = 2$ and $s = 0, 1, 2, \dots$. Thus, $\sigma(t)$ is in the form of (1) with $\omega_1 = 0.5$, $\omega_2 = 0.3$, and $\omega_3 = 0.2$. The three digraphs $\mathcal{G}_i$, $i = 1, 2, 3$, are shown in Fig. 1. It can be seen that Assumptions 2.1 and 2.2 are satisfied even though $\mathcal{G}_{\sigma(t)}$ is disconnected at any time $t \geq 0$. We let $a_{ij}(t) = 1$ whenever $(j, i) \in \mathcal{E}_{\sigma(t)}$.

The three matrices associated with the three digraphs $\mathcal{G}_i$, $i = 1, 2, 3$, are given by $H_1 = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $H_2 =$

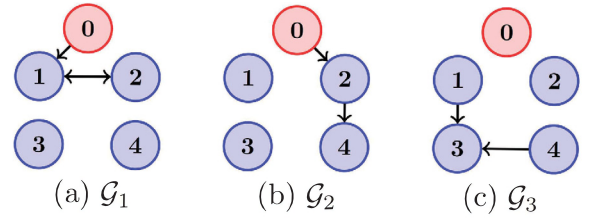


Fig. 1. Communication graphs (a) $\mathcal{G}_1$, (b) $\mathcal{G}_2$, and (c) $\mathcal{G}_3$.

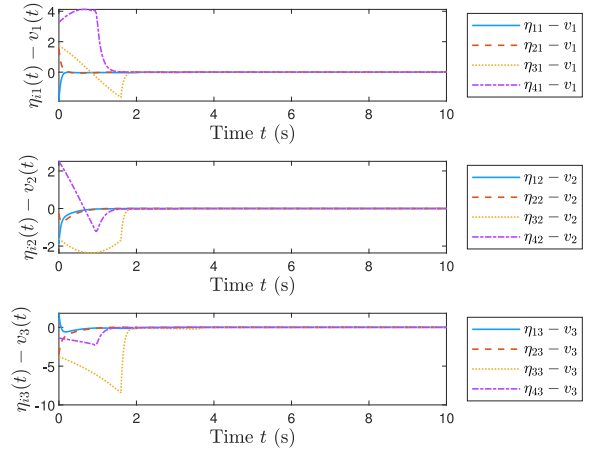


Fig. 2. Estimation errors of the leader's state v .

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \text{ and } H_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ From (7), } \bar{\mu}_1 = \frac{5}{3}.$$

According to Theorem 3.1, we can design a distributed dynamic measurement output feedback control law of the form (37) with the following design parameters: $\mu_1 = 8$, $L_i = \text{col}(-14, -48)$, $K_{1i} = [-8 \ -6]$, and $K_{2i} = U_i - K_{1i} X_i = [7 \ 6 \ -0.5 * i]$, $i = 1, 2, 3, 4$. It can be verified that $A_i + B_i K_{1i}$ and $A_i + L_i C_{mi}$, $i = 1, 2, 3, 4$, are Hurwitz.

Since the largest magnitude of the eigenvalues of the state transition matrix $\Phi(2, 0)$ of system (22) for this example is 0.0361, $\Phi(2, 0)$ is Schur. Thus, by Lemma 2.4, the origin of system (22) for this example is exponentially stable. An estimated upper bound for the period computed using Lemma 2.5 is $\bar{T} = 0.001$, which indeed confirms the conservativeness of Lemma 2.5. Thus, instead of using Lemma 2.5, we use the alternative approach to determine T as described in Remark 2.2.

Our simulation is conducted with $v_1(0) = v_2(0) = v_3(0) = 2$ and other initial conditions randomly chosen. The estimation errors of the distributed observer and the error outputs of the followers are shown in Figs. 2 and 3, respectively.

It is noted that, since S has an eigenvalue with a positive real part, the problem in this section cannot be handled by the distributed observer in Su and Huang (2012b).

5. Conclusion

In this paper, after establishing a stability result for a class of linear periodic switched systems, we have developed a distributed observer and an adaptive distributed observer for general linear leader systems over periodic jointly connected switching communication networks. As an application of the distributed observer, we have further solved the cooperative output regulation

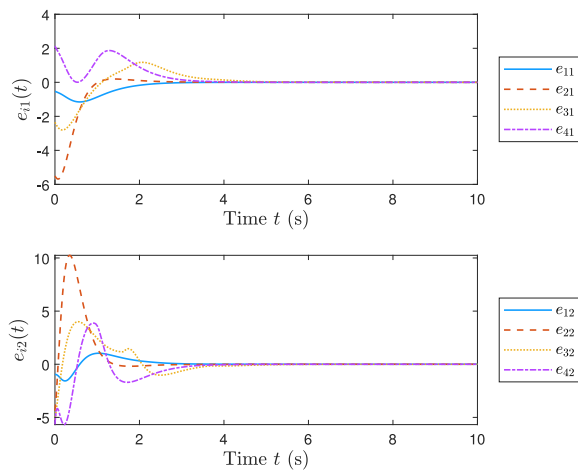


Fig. 3. Error outputs $e_i = \text{col}(e_{i1}, e_{i2})$ of the followers.

problem for linear multi-agent systems with general linear leader systems over periodic jointly connected switching networks.

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